Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Shiky
6/3/26

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Session	**2024-25 (EVEN)**	**Course Name**	**Mobile and Web Application for IOT Lab**
Semester	**6**	**Course Code**	**23IOT1624**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**07**
Course Outcome	1. Understand and apply basic Kotlin programming constructs including variables, control structures, functions, null safety, and object-oriented concepts for Android app development. 2. Design and develop responsive user interfaces using XML layout files, ViewGroups, and widgets in Android Studio, adhering to Material Design principles. 3. Implement event handling and application logic using Kotlin to capture user input, manage app state, navigate between screens, and interact with Android system components like Toasts and Dialogs 4. Integrate learned concepts to develop a mini Android application that demonstrates use of ViewModel, multi-screen navigation, and proper UI/UX design for real-world problem-solving
Aim	Capture user input and handle UI events with navigation.
Problem Definition	Button click listeners, Toast/Snackbar, activity navigation using Intent, data passing.
Theory (100 words)	In Android application development, user interaction is handled through graphical user interface (UI) components such as **EditText, Button, TextView, Toast, and Snackbar**. These components allow users to enter data and perform actions within the application. The **EditText** widget is used to capture user input such as name, email, phone number, and city. A **Button** is used to trigger events when the user clicks it. Event handling in Android is implemented using **click listeners**, which detect user actions on UI elements. When the button is clicked, the application processes the input data and performs operations such as validation and displaying messages. **Toast messages** provide short notifications on the screen, while **Snackbar messages** display brief feedback at the bottom of the interface. Android applications can navigate between different screens using **Activities**. Navigation between activities is achieved through **Intents**, which are messaging objects used to start another activity and transfer data between them. In this practical, user data entered in the first activity is passed to the second activity using an **Intent**, where it is displayed using a

	TextView. This demonstrates basic user input handling, event management, and activity navigation in Android applications.
Procedure and Execution (100 Words)	**Steps for Implementation** 1. Open **Android Studio** and create a new Android project using the **Empty Activity** template. 2. Enter the required details such as **application name, package name, language (Kotlin), and minimum SDK version**, then finish creating the project. 3. Navigate to the **res → layout** folder and open the **activity_main.xml** file. 4. Add a **LinearLayout** as the main layout container to arrange the user interface elements vertically. 5. Drag and place **EditText fields** in the layout to accept user inputs such as **Name, Email, Phone, and City**. 6. Add a **Button** widget below the input fields to allow the user to submit the entered data. 7. Set appropriate properties for each widget such as **id, hint, width, height, margin, and alignment** in the XML file. 8. Open the **MainActivity.kt** file and connect the UI components using **findViewById()** to access the input fields and button in the activity. 9. Implement a **button click listener** to capture the user input and validate whether all fields are filled. 10. Display feedback messages using **Toast or Snackbar** to inform the user about the entered data or missing inputs. 11. Create another activity named **SecondActivity** to display the submitted information. 12. Use an **Intent** to navigate from **MainActivity to SecondActivity** and pass the user input data between activities. 13. In **SecondActivity**, receive the data using intent.getStringExtra() and display it using a **TextView**. 14. Add the new activity in the **AndroidManifest.xml** file if it is not automatically added. 15. Save all files, **build the project**, and run the application on an **Android emulator or physical device** to test the

functionality of user input, event handling, and activity navigation.

Code:

1. Manifest

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools">

   <application
      android:allowBackup="true"
      android:dataExtractionRules="@xml/data_extraction_rules"
      android:fullBackupContent="@xml/backup_rules"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"
      android:roundIcon="@mipmap/ic_launcher_round"
      android:supportsRtl="true"
      android:theme="@style/Theme.Practical7_iot63">

      <activity
         android:name=".SecondActivity"
         android:exported="false" />

      <activity
         android:name=".MainActivity"
         android:exported="true">

         <intent-filter>
            <action android:name="android.intent.action.MAIN" />

            <category android:name="android.intent.category.LAUNCHER"
/>
         </intent-filter>

      </activity>

   </application>

</manifest>
```

2. MainActivity.kt

```
package com.example.practical7_iot63

import android.content.Intent
```

```
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.*
import com.google.android.material.snackbar.Snackbar

class MainActivity : AppCompatActivity() {

    lateinit var nameInput: EditText
    lateinit var emailInput: EditText
    lateinit var phoneInput: EditText
    lateinit var cityInput: EditText
    lateinit var submitBtn: Button

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        nameInput = findViewById(R.id.nameInput)
        emailInput = findViewById(R.id.emailInput)
        phoneInput = findViewById(R.id.phoneInput)
        cityInput = findViewById(R.id.cityInput)
        submitBtn = findViewById(R.id.submitBtn)

        submitBtn.setOnClickListener {

            val name = nameInput.text.toString()
            val email = emailInput.text.toString()
            val phone = phoneInput.text.toString()
            val city = cityInput.text.toString()

            if (name.isEmpty() || email.isEmpty() || phone.isEmpty() ||
city.isEmpty()) {

                Toast.makeText(this, "Please fill all fields",
Toast.LENGTH_SHORT).show()

            } else {

                Snackbar.make(it, "Data Submitted",
Snackbar.LENGTH_SHORT).show()

                val intent = Intent(this, SecondActivity::class.java)
```

```
            intent.putExtra("name", name)
            intent.putExtra("email", email)
            intent.putExtra("phone", phone)
            intent.putExtra("city", city)

            startActivity(intent)
        }
    }
  }
}
```

3. ActivityMain.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  android:orientation="vertical"
  android:padding="20dp"
  android:gravity="center"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <EditText
    android:id="@+id/nameInput"
    android:hint="Enter Name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

  <EditText
    android:id="@+id/emailInput"
    android:hint="Enter Email"
    android:layout_marginTop="10dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

  <EditText
    android:id="@+id/phoneInput"
    android:hint="Enter Phone"
    android:layout_marginTop="10dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

  <EditText
    android:id="@+id/cityInput"
    android:hint="Enter City"
    android:layout_marginTop="10dp"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/submitBtn"
        android:text="Submit"
        android:layout_marginTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

4. SecondActivity.kt

```
package com.example.practical7_iot63

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.TextView

class SecondActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_second)

        val result = findViewById<TextView>(R.id.resultText)

        val name = intent.getStringExtra("name")
        val email = intent.getStringExtra("email")
        val phone = intent.getStringExtra("phone")
        val city = intent.getStringExtra("city")

        result.text = """
Name: $name
Email: $email
Phone: $phone
City: $city
        """.trimIndent()
    }
}
```

5. Activity_second.xml

```
<LinearLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:gravity="center"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TextView
      android:id="@+id/resultText"
      android:textSize="20sp"
      android:textStyle="bold"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"/>

</LinearLayout>
```

Output:

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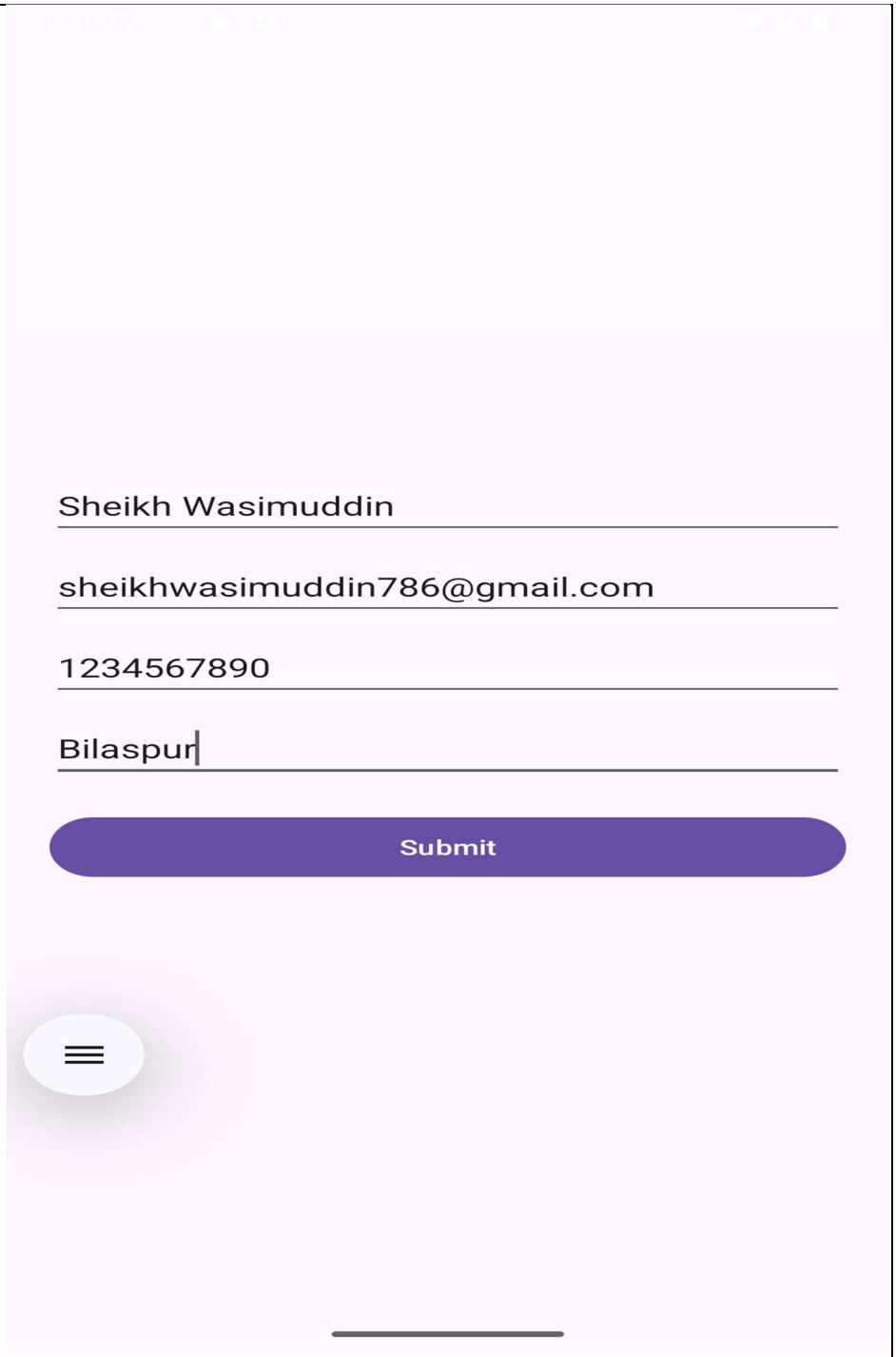

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Email: sheikhwasimuddin786@gmail.com
Phone: 1234567890
City: Bilaspur

Output Analysis	When the application is executed, the first screen displays a user interface containing four input fields: Name, Email, Phone, and City, along with a Submit button. The user enters the required information into the respective EditText fields. When the user clicks the Submit button, the application captures the entered data using the button click listener defined in the activity code. If any of the input fields are left empty, the application displays a Toast message prompting the user to fill all the fields. If all the inputs are provided correctly, a Snackbar message appears confirming that the data has been submitted successfully. After validation, the application uses an Intent to navigate from MainActivity to SecondActivity while passing the entered data. The second screen then retrieves the data and displays the information using a TextView. This output confirms that the application correctly handles user input, event handling, message display, and activity navigation with data passing between activities.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	In this practical, an Android application was successfully developed to capture user input and handle user interface events. The application uses UI components such as EditText, Button, Toast, and Snackbar to interact with the user. The button click listener was implemented to process user actions and validate the entered information. Activity navigation was achieved using Intent, which allowed the application to move from one screen to another while passing user data. The entered details were displayed in the second activity, demonstrating successful data transfer between activities. This practical helps in understanding the basic concepts of user input handling, event management, and activity navigation in Android applications.

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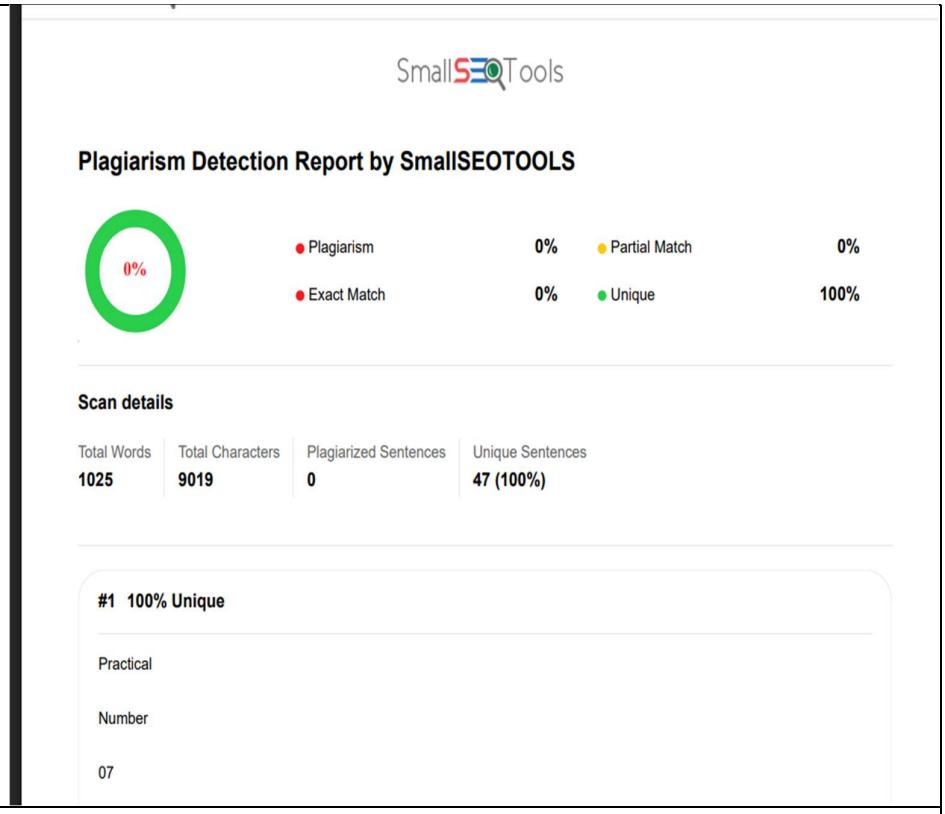
Plag Report (Similarity index < 12%)	
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Practical

Number

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